

Theorem 9.1.3. *For all $k, c, r \geq 1$ there exists an $n \geq k$ such that every n -set X has a monochromatic r -subset with respect to any c -colouring of $[X]^k$.* [9.3.3]

Proof. As is customary in set theory, we denote by $n \in \mathbb{N}$ (also) the set $\{0, \dots, n-1\}$. Suppose the assertion fails for some k, c, r . Then for every $n \geq k$ there exist an n -set, without loss of generality the set n , and a c -colouring $[n]^k \rightarrow c$ such that n contains no monochromatic r -set. Let us call such colourings *bad*; we are thus assuming that for every $n \geq k$ there exists a bad colouring of $[n]^k$. Our aim is to combine these into a bad colouring of $[\mathbb{N}]^k$, which will contradict Theorem 9.1.2. (8.1.2)
 k, c, r

For every $n \geq k$ let $V_n \neq \emptyset$ be the set of bad colourings of $[n]^k$. For $n > k$, the restriction $f(g)$ of any $g \in V_n$ to $[n-1]^k$ is still bad, and hence lies in V_{n-1} . By the infinity lemma (8.1.2), there is an infinite sequence $g_k, g_{k+1}, \dots$ of bad colourings $g_n \in V_n$ such that $f(g_n) = g_{n-1}$ for all $n > k$. For every $m \geq k$, all colourings g_n with $n \geq m$ agree on $[m]^k$, so for each $Y \in [\mathbb{N}]^k$ the value of $g_n(Y)$ coincides for all $n > \max Y$. Let us define $g(Y)$ as this common value $g_n(Y)$. Then g is a bad colouring of $[\mathbb{N}]^k$: every r -set $S \subseteq \mathbb{N}$ is contained in some sufficiently large n , so S cannot be monochromatic since g coincides on $[n]^k$ with the bad colouring g_n . $\square$

The least integer n associated with k, c, r as in Theorem 9.1.3 is the *Ramsey number* for these parameters; we denote it by $R(k, c, r)$. Ramsey number
 $R(k, c, r)$

9.2 Ramsey numbers

Ramsey's theorem may be rephrased as follows: if $H = K^r$ and G is a graph with sufficiently many vertices, then either G itself or its complement $\overline{G}$ contains a copy of H as a subgraph. Clearly, the same is true for any graph H , simply because $H \subseteq K^h$ for $h := |H|$.

However, if we ask for the *least* n such that every graph G of order n has the above property – this is the *Ramsey number* $R(H)$ of H – then the above question makes sense: if H has only few edges, it should embed more easily in G or $\overline{G}$, and we would expect $R(H)$ to be smaller than the Ramsey number $R(h) = R(K^h)$. Ramsey number
 $R(H)$

A little more generally, let $R(H_1, H_2)$ denote the least $n \in \mathbb{N}$ such that $H_1 \subseteq G$ or $H_2 \subseteq \overline{G}$ for every graph G of order n . For most graphs H_1, H_2 , only very rough estimates are known for $R(H_1, H_2)$. Interestingly, lower bounds given by random graphs (as in Theorem 11.1.3) are often sharper than even the best bounds provided by explicit constructions. $R(H_1, H_2)$

The following proposition describes one of the few cases where exact Ramsey numbers are known for a relatively large class of graphs:

Proposition 9.2.1. *Let s, t be positive integers, and let T be a tree of order t . Then $R(T, K^s) = (s-1)(t-1) + 1$.*

(1.5.3) *Proof.* The disjoint union of $s-1$ graphs K^{t-1} contains no copy of T , while the complement of this graph, the complete $(s-1)$ -partite graph K_{t-1}^{s-1} , does not contain K^s . This proves $R(T, K^s) \geq (s-1)(t-1) + 1$.
(5.2.3)

Conversely, let G be any graph of order $n = (s-1)(t-1) + 1$ whose complement contains no K^s . Then $s > 1$, and in any vertex colouring of G (in the sense of Chapter 5) at most $s-1$ vertices can have the same colour. Hence, $\chi(G) \geq \lceil n/(s-1) \rceil = t$. By Lemma 5.2.3, G has a subgraph H with $\delta(H) \geq t-1$, which by Corollary 1.5.3 contains a copy of T . $\square$

As the main result of this section, we shall now prove one of those rare general theorems providing a relatively good upper bound for the Ramsey numbers of a large class of graphs, a class defined in terms of a standard graph invariant. The theorem deals with the Ramsey numbers of sparse graphs: it says that the Ramsey number of graphs H with bounded maximum degree grows only linearly in $|H|$ – an enormous improvement on the exponential bound from the proof of Theorem 9.1.1.

Theorem 9.2.2. (Chvátal, Rödl, Szemerédi & Trotter 1983)

For every positive integer Δ there is a constant c such that

$$R(H) \leq c|H|$$

for all graphs H with $\Delta(H) \leq \Delta$.

(7.1.1) *Proof.* The basic idea of the proof is as follows. We wish to show that
(7.4.1) $H \subseteq G$ or $H \subseteq \bar{G}$ if $|G|$ is large enough (though not too large). Consider
(7.5.2) an ϵ -regular partition of G , as provided by the regularity lemma. If
(9.1.1) enough of the ϵ -regular pairs in this partition have high density, we may hope to find a copy of H in G . If most pairs have low density, we try to find H in $\bar{G}$. Let R , R' and R'' be the regularity graphs of G whose edges correspond to the pairs of density ≥ 0 ; $\geq 1/2$; $< 1/2$ respectively.³ Then R is the edge-disjoint union of R' and R'' .

Now to obtain $H \subseteq G$ or $H \subseteq \bar{G}$, it suffices by the blow-up lemma, Lemma 7.5.2, to ensure that H is contained in a suitable ‘inflated regularity graph’ R'_s or R''_s . Since $\chi(H) \leq \Delta(H) + 1 \leq \Delta + 1$, this will be the case if $s \geq \alpha(H)$ and we can find a $K^{\Delta+1}$ in R' or in R'' . But that is easy to ensure: we just need that $K^r \subseteq R$, where r is the Ramsey number of $\Delta + 1$, which will follow from Turán’s theorem because R is dense.

For the formal proof let now $\Delta \geq 1$ be given. On input $d := 1/2$ and Δ , Lemma 7.5.2 returns an ϵ_0 . Let $m := R(\Delta + 1)$ be the Ramsey number of $\Delta + 1$. Let $\epsilon \leq \epsilon_0$ be positive but small enough that for $k = m$

³ In our formal proof later we shall define R'' a little differently, so that it complies properly with our definition of a regularity graph.

Δ, d
 ϵ_0, m
 ϵ

(and hence for all $k \geq m$)

$$2\epsilon < \frac{1}{m-1} - \frac{1}{k}; \quad (1)$$

then in particular $\epsilon < 1$. Finally, let M be the integer returned by the regularity lemma (Theorem 7.4.1) on input ϵ and m . M

All the quantities defined so far depend only on Δ . We shall prove the theorem with

$$c := \frac{2^{\Delta+1}M}{1-\epsilon}.$$
 c

Let H with $\Delta(H) \leq \Delta$ be given, and let $s := |H|$. Let G be an arbitrary graph of order $n \geq c|H|$; we show that $H \subseteq G$ or $H \subseteq \overline{G}$. s
 G, n

By Theorem 7.4.1, G has an ϵ -regular partition $\{V_0, V_1, \dots, V_k\}$ with exceptional set V_0 and $|V_1| = \dots = |V_k| =: \ell$, where $m \leq k \leq M$. Then k
 ℓ

$$\ell = \frac{n - |V_0|}{k} \geq n \frac{1-\epsilon}{M} \geq cs \frac{1-\epsilon}{M} \geq 2^{\Delta+1}s = 2s/d^\Delta. \quad (2)$$

Let R be the regularity graph with parameters $\epsilon, \ell, 0$ corresponding to this partition. By definition, R has k vertices and R

$$\begin{aligned} \|R\| &\geq \binom{k}{2} - \epsilon k^2 \\ &= \frac{1}{2}k^2 \left(1 - \frac{1}{k} - 2\epsilon\right) \\ &\stackrel{(1)}{>} \frac{1}{2}k^2 \left(1 - \frac{1}{k} - \frac{1}{m-1} + \frac{1}{k}\right) \\ &= \frac{1}{2}k^2 \frac{m-2}{m-1} \\ &\geq t_{m-1}(k) \end{aligned}$$

edges. By Theorem 7.1.1, therefore, R has a subgraph $K = K^m$. K

We now colour the edges ij of R with two colours: red if the pair (V_i, V_j) has density at least $1/2$, and green otherwise. Let R' be the spanning subgraph of R formed by the red edges, and R'' the spanning subgraph of R formed by the green edges and those whose corresponding pair has density exactly $1/2$. Then R' is a regularity graph of G with parameters ϵ, ℓ and $1/2$. And R'' is a regularity graph of $\overline{G}$, with the same parameters: as one easily checks, every pair (V_i, V_j) that is ϵ -regular for G is also ϵ -regular for $\overline{G}$.

By definition of m , our graph K contains a red or a green K^r , for $r := \chi(H) \leq \Delta + 1$. Correspondingly, $H \subseteq R'_s$ or $H \subseteq R''_s$. Since $\epsilon \leq \epsilon_0$ and $\ell \geq 2s/d^\Delta$ by (2), both R' and R'' satisfy the requirements of Lemma 7.5.2, so $H \subseteq G$ or $H \subseteq \overline{G}$ as desired. $\square$

Ramsey-
minimal

So far in this section, we have been asking what is the least order of a complete graph G such that every 2-colouring of its edges yields a monochromatic copy of some given graph H . Rather than keeping G complete and focusing on its order, let us now consider its structure too, i.e., minimize G with respect to the subgraph relation. Given a graph H , let us call a graph G *Ramsey-minimal* for H if G is minimal with the property that every 2-colouring of its edges yields a monochromatic copy of H .

What do such Ramsey-minimal graphs look like? Are they unique? The following result, which we include for its pretty proof, answers the second question for some H :

Proposition 9.2.3. *If T is a tree but not a star, then infinitely many graphs are Ramsey-minimal for T .*

(1.5.3)
(5.2.3)
(5.2.5)
(=11.2.2)

Proof. Let $|T| =: r$. We show that for every $n \in \mathbb{N}$ there is a graph of order at least n that is Ramsey-minimal for T .

By Theorem 5.2.5, there exists a graph G with chromatic number $\chi(G) > r^2$ and girth $g(G) > n$. If we colour the edges of G red and green, then the red and the green subgraph cannot both have an r -(vertex-)colouring in the sense of Chapter 5: otherwise we could colour the vertices of G with the pairs of colours from those colourings and obtain a contradiction to $\chi(G) > r^2$. So let $G' \subseteq G$ be monochromatic with $\chi(G') > r$. By Lemma 5.2.3, G' has a subgraph of minimum degree at least r , which contains a copy of T by Corollary 1.5.3.

Let $G^* \subseteq G$ be Ramsey-minimal for T . Clearly, G^* is not a forest: the edges of any forest can be 2-coloured (partitioned) so that no monochromatic subforest contains a path of length 3, let alone a copy of T . (Here we use that T is not a star, and hence contains a P^3 .) So G^* contains a cycle, which has length $g(G) > n$ since $G^* \subseteq G$. In particular, $|G^*| > n$ as desired. $\square$

9.3 Induced Ramsey theorems

Ramsey's theorem can be rephrased as follows. For every graph $H = K^r$ there exists a graph G such that every 2-colouring of the edges of G yields a monochromatic $H \subseteq G$. This is witnessed by any large enough complete graph as G . Let us now change the problem slightly and ask for a graph G in which every 2-edge-colouring yields a monochromatic induced $H \subseteq G$, where H is now an arbitrary given graph.

This slight modification changes the character of the problem. What is needed now is no longer a simple proof that G is 'big enough' (as for Theorem 9.1.1), but a careful construction: the construction of a graph that, however we bipartition its edge set, contains an induced copy of H